

DRAGASANI, RomaniaWMO: **153950**Lat: **44.6656N**Lon: **24.2372E**Elev: **281**StdP: **97.99**Time Zone: **2.00 (EUE)**Period: **86-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.7	-7.8	-12.1	1.4	-7.4	-10.2	1.6	-5.3	9.5	4.6	8.4	3.7	1.7	20	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.5	33.0	20.6	31.7	20.4	30.1	20.0	22.9	29.2	21.9	28.4	21.1	27.8	2.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	16.1	26.7	19.8	15.0	25.1	18.9	14.2	24.1	69.2	29.3	65.2	28.6	62.2	27.7	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	7.9	6.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	-0.5	3.3	7.1	11.9	17.2	21.4	23.0	23.7	18.4	12.3	6.3	2.0
	DBStd	9.17	4.34	4.85	4.98	4.66	3.15	2.97	2.83	3.17	4.00	4.42	4.59	4.16
	HDD10.0	1074	326	191	117	34	1	0	0	0	0	28	128	249
	HDD18.3	2727	584	421	349	198	57	7	2	3	48	189	361	507
	CDD10.0	1889	0	3	26	91	226	343	404	426	253	100	17	1
	CDD18.3	501	0	0	0	3	23	101	147	171	52	3	0	0
	CDH23.3	4326	0	0	1	30	163	818	1312	1582	394	26	0	0
	CDH26.7	1527	0	0	0	2	18	251	482	662	111	1	0	0
Wind	WSAvg	2.4	2.1	2.7	2.9	3.2	2.9	2.4	2.2	2.1	2.5	2.1	2.1	2.2
Precipitation	PrecAvg	616	37	30	37	53	70	70	74	49	55	52	40	45
	PrecMax	1128	73	64	115	147	160	140	200	135	130	125	108	143
	PrecMin	325	6	4	4	15	16	16	19	0	2	0	0	0
	PrecStd	165	21	18	29	27	36	32	46	30	38	31	26	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	17.8	22.6	26.4	28.7	33.2	34.2	35.1	31.9	26.1	19.1	14.2
		MCWB	7.9	10.5	13.6	16.0	17.8	21.1	22.1	21.1	19.7	16.1	13.6	10.2
	2%	DB	9.1	14.4	19.5	23.7	26.6	31.2	32.6	33.3	29.6	23.5	16.6	11.6
		MCWB	6.0	9.0	11.6	15.4	17.6	20.7	21.0	20.5	18.0	15.2	12.0	8.1
	5%	DB	7.1	12.0	16.9	21.4	25.2	29.4	31.1	32.0	27.4	21.4	14.3	9.5
		MCWB	4.7	7.4	10.4	13.8	17.1	20.4	20.6	20.3	17.7	14.3	11.0	7.1
	10%	DB	5.3	9.9	14.6	19.2	23.5	27.7	29.4	30.4	25.3	19.1	12.3	7.8
		MCWB	3.5	6.5	9.2	12.6	16.1	19.5	20.2	20.0	17.2	13.5	10.2	5.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.2	11.7	16.0	17.1	20.4	23.0	24.6	24.6	21.2	17.1	14.1	11.4
		MCDB	11.2	15.9	19.2	24.4	26.4	30.4	31.4	31.9	28.0	22.2	17.6	13.4
	2%	WB	6.6	9.7	12.4	15.7	18.7	22.0	22.9	23.2	19.7	16.0	13.0	8.8
		MCDB	8.8	14.1	17.4	22.5	24.2	29.0	28.6	29.7	26.3	20.9	15.4	10.6
	5%	WB	5.1	8.1	10.8	14.7	17.8	21.1	21.8	21.7	18.6	15.2	11.8	7.2
		MCDB	6.5	11.3	16.0	20.8	23.0	27.8	28.1	28.3	24.7	20.2	13.9	9.5
	10%	WB	3.6	6.8	9.5	13.3	16.9	20.3	20.9	20.9	17.8	14.2	10.5	5.8
		MCDB	5.1	9.2	13.9	18.3	22.0	26.8	27.7	27.7	24.0	18.5	12.1	7.7
Mean Daily Temperature Range	5% DB	MDBR	6.0	7.4	8.7	9.4	10.0	10.5	10.8	11.5	10.1	8.8	6.1	6.0
		MCDBR	8.8	10.4	12.6	12.2	12.0	12.2	12.4	12.9	12.6	12.4	9.4	8.5
		MCWBR	5.8	6.4	6.5	6.3	5.1	4.9	4.3	4.1	4.4	5.6	5.3	5.8
	5% WB	MCDBR	7.4	10.1	11.6	11.6	11.0	11.7	11.5	11.7	10.7	10.9	7.7	8.2
		MCWBR	5.3	6.4	6.4	6.2	5.6	5.2	4.8	4.8	4.6	5.1	4.5	5.8
Clear-Sky Solar Irradiance	taub	0.324	0.355	0.389	0.445	0.424	0.428	0.440	0.458	0.417	0.381	0.356	0.328	
	taud	2.332	2.259	2.232	2.089	2.216	2.240	2.212	2.156	2.238	2.334	2.351	2.356	
	Ebn at Noon	778	816	837	816	848	845	828	795	798	778	732	732	
	Edh at Noon	80	103	122	152	138	136	138	141	120	95	79	72	
All-Sky Solar Radiation	RadAvg	1.42	2.26	3.46	4.68	5.75	6.35	6.38	5.63	4.08	2.67	1.50	1.15	
	RadStd	0.20	0.35	0.40	0.46	0.44	0.47	0.43	0.41	0.34	0.35	0.20	0.16	